CHAPTER 1

INTRODUCTION

1.1 Introduction

Study Sync is a web-based personal study planner developed using the **MERN Stack**. It helps students organize their subjects, create study tasks, manage study time, and monitor their academic progress in one application. Students often have difficulty managing multiple subjects, assignments, deadlines and daily study hours. Study Sync provides a centralized platform where students can set their daily study goals, add subjects with deadlines, create study tasks, use a study timer and monitor their overall progress. The application provides a dashboard that gives a quick overview of the student's current study activities. It displays the daily study goal, total subjects, study time, day streak and recommended study plan. The system uses **React.js** for the frontend, **Node.js and Express.js** for the backend, and **MongoDB** for storing application data.

1.2 System Overview

- User Authentication – Allows users to register and securely access their study account.
- Dashboard – Displays daily study goals, subjects, study time, streaks and overall progress.
- My Subjects – Allows users to add and manage subjects, target study hours and deadlines.
- Study Tasks – Allows users to create, manage and track study tasks.
- Study Timer & Progress – Helps users conduct focused study sessions and monitor their study progress.

User Registration → Dashboard → Add Subjects → Create Study Tasks → Start Study Session → Save Study Time → Track Progress

The frontend of the application is developed using React.js, while Node.js and Express.js are used to develop the backend REST APIs. MongoDB is used to store user, subject, task and study-session information.

CHAPTER 2

METHODOLOGY

2.1 Objective

The main objective of StudySync is to develop a web-based personal study planner that helps students organize and monitor their study activities. The system combines subject management, task management, study-time tracking and progress monitoring into a single application.

The objectives of the project are:

- To provide a centralized platform for managing study activities.
- To allow students to add and manage their subjects.
- To help students set daily study goals and subject-wise study targets.
- To allow students to create and manage study tasks.
- To provide a study timer for focused study sessions.
- To record completed study sessions and study time.
- To monitor subject-wise study progress.
- To provide a dashboard for viewing important study information.
- To store user and study-related information using MongoDB.

2.2 Problem Statement

Students commonly use separate methods such as notebooks, mobile reminders and simple timers to manage their studies. These methods do not provide a single platform for managing subjects, tasks, deadlines and study time.

The major problems are:

- Difficulty in organizing multiple subjects.
- Lack of proper tracking of daily study hours.
- Difficulty in remembering upcoming deadlines.
- No centralized task management.

- Lack of detailed study progress tracking.
- Difficulty in maintaining a record of completed study sessions.

Therefore, there is a need for a web application that combines these activities into a single study-management system.

2.3 Project Description

StudySync is a web-based personal study planner developed using the MERN Stack. It helps students manage their subjects, study tasks, study time and academic progress through a single platform.The system provides a Dashboard to view daily study goals, total subjects, study time, day streak and overall progress. The My Subjects module allows users to add subjects, set target study hours and deadlines. The Study Tasks module helps users create and manage tasks based on their study requirements.

The Study Timer allows users to conduct focused study sessions by selecting a subject and task. Completed sessions are recorded and used to calculate study progress. The My Progress module displays total study time, completed sessions and subject-wise progress.The frontend is developed using React.js, while Node.js and Express.js are used for backend REST APIs. MongoDB is used to store user, subject, task and study-session information.

CHAPTER 3

RESULTS AND DISCUSSION

Screenshots

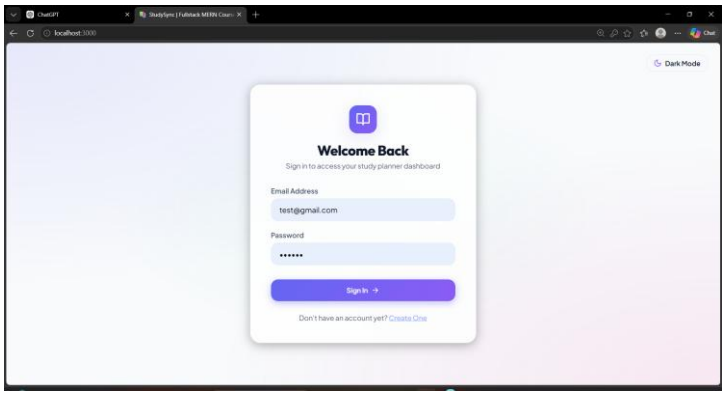


Figure 3.1 – Login

Allows students to log in securely using their email and password.

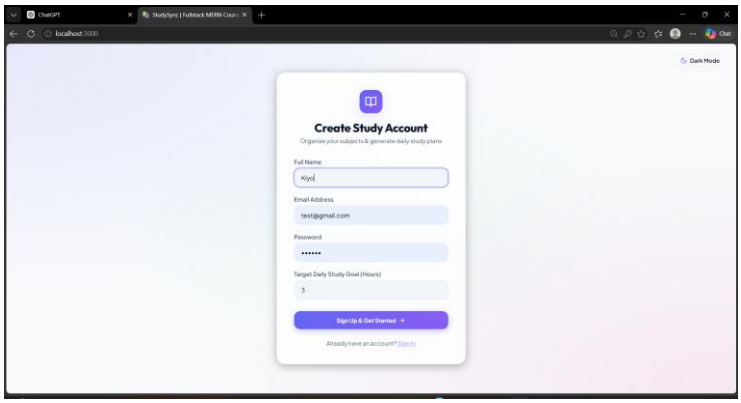


Figure 3.2 – Register

Allows new students to create a Study Planner account by entering their details.

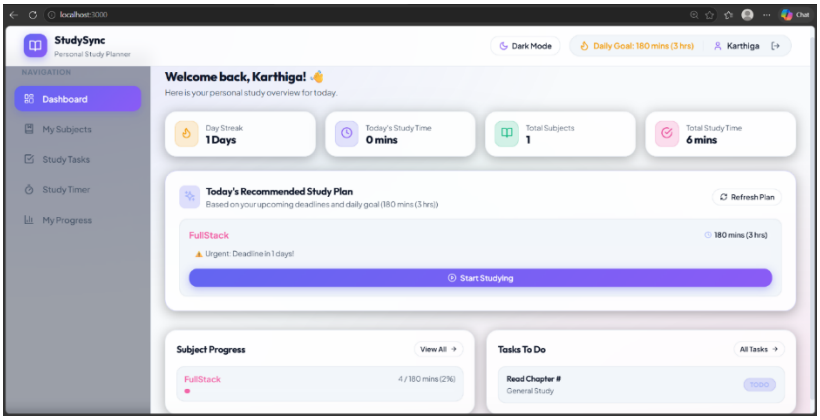


Figure 3.3 – Dashboard

Shows the user's study overview, goals, subjects and progress.

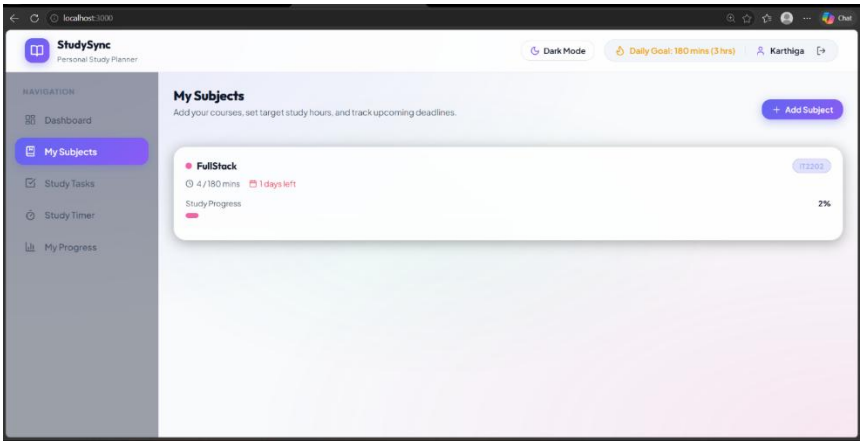


Figure 3.4 – My Subjects

Displays subjects, study targets, deadlines and progress.

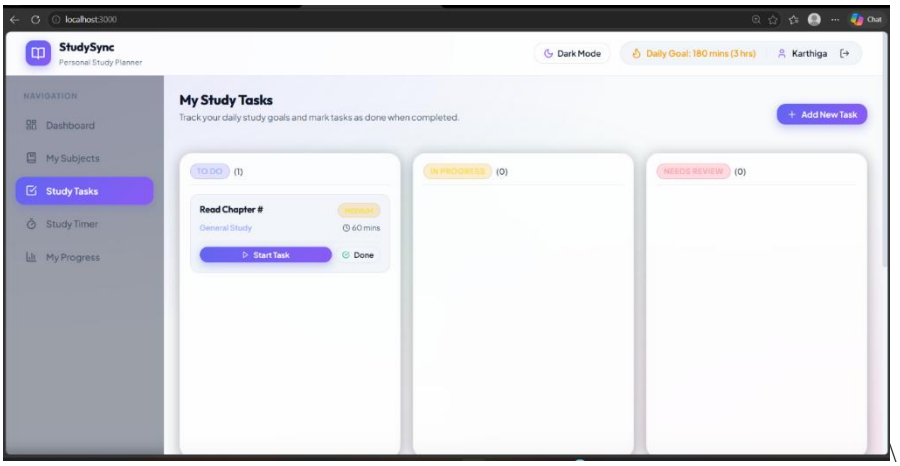


Figure 3.5 – Study Tasks

Allows users to create and manage study tasks.

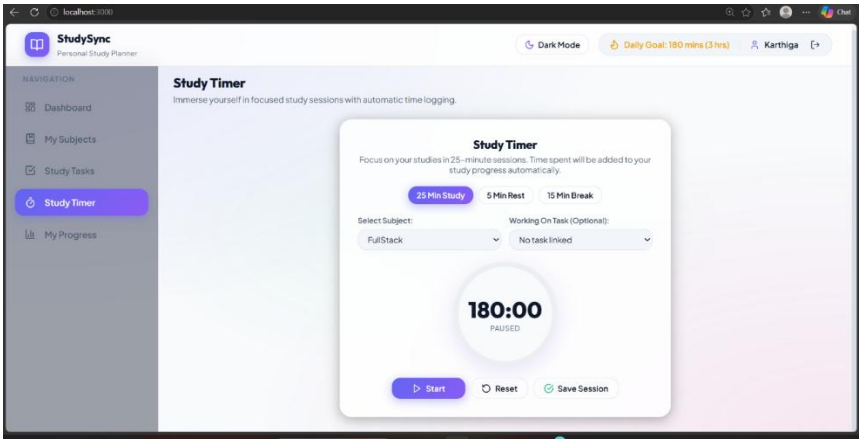


Figure 3.6 – Study Timer

Provides a timer for focused study sessions.

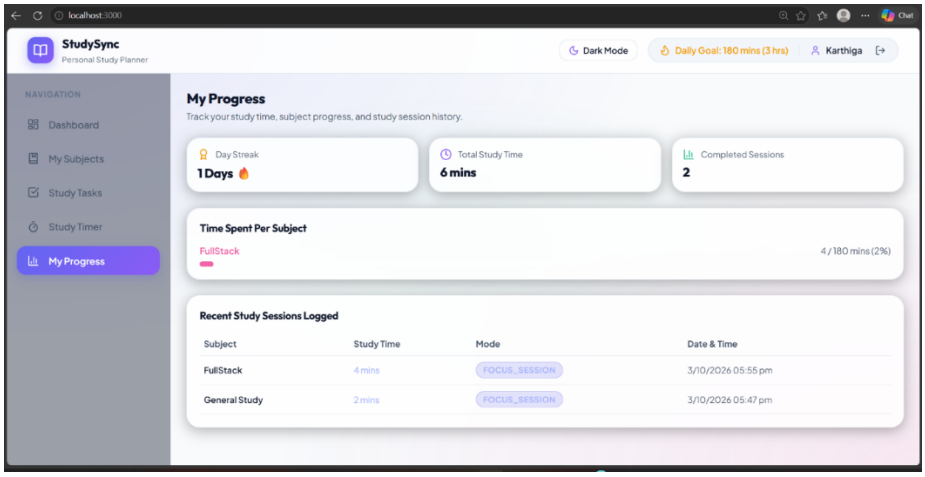


Figure 3.7 – My Progress

Displays study time, completed sessions and subject progress.

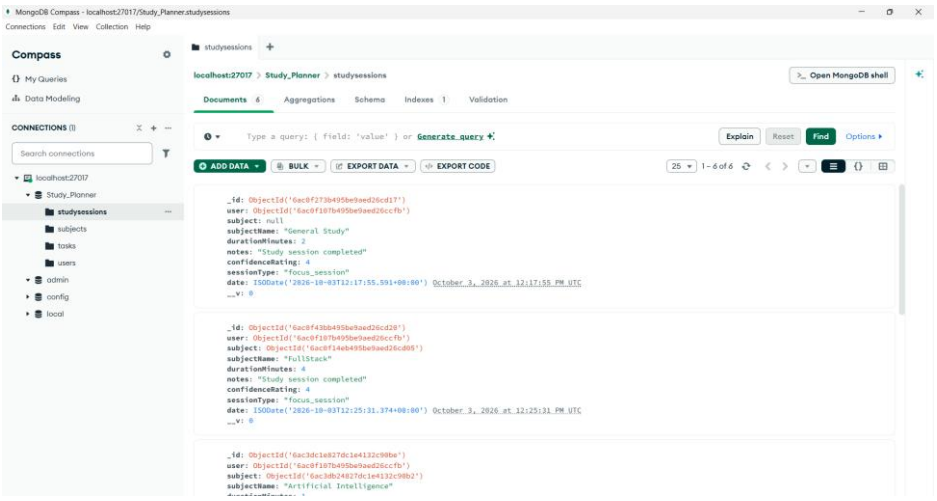


Figure 3.8- Study Sessions Collection

Shows stored study session details in MongoDB.

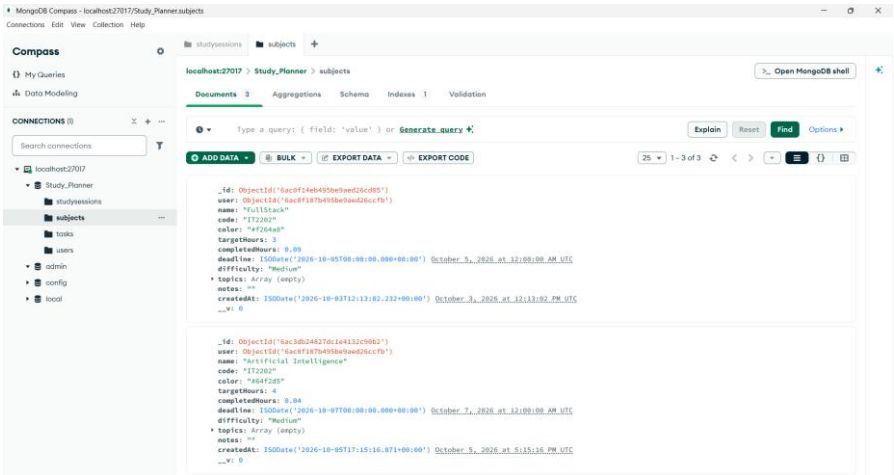


Figure 3.9 – Subjects Collection

Shows the subjects added and stored in MongoDB.

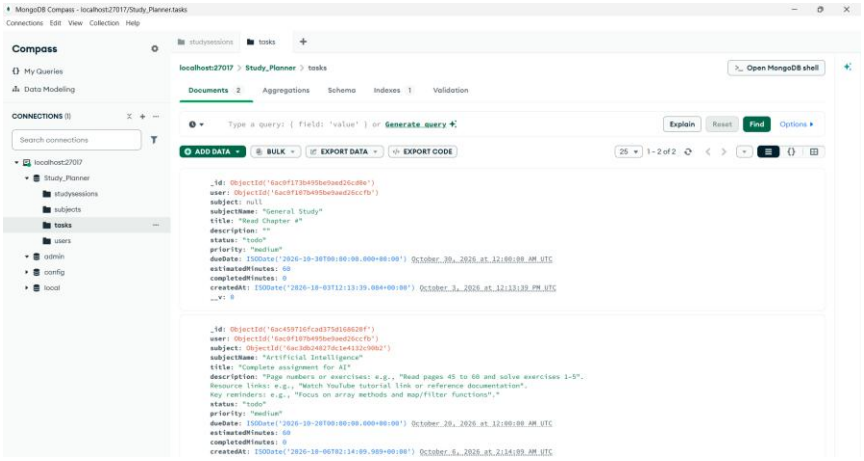


Figure 3.10 – Tasks Collection

Shows the tasks added and stored in MongoDB.

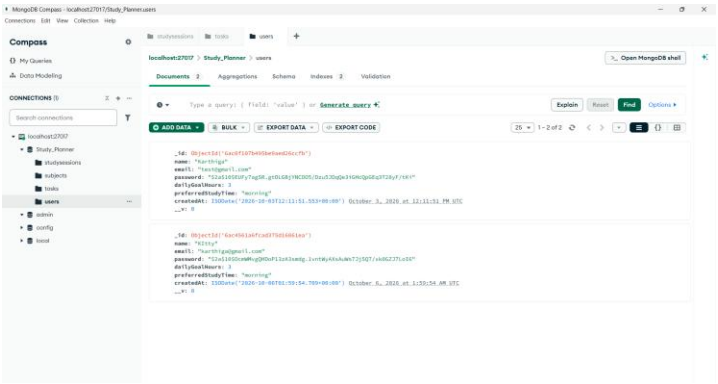


Figure 3.11 – Users Collection

Shows registered user details stored in MongoDB.

Github Link:

https://github.com/Karthi-Shadow/StudySync

CHAPTER 4

CONCLUSION

StudySync successfully provides a simple and organized platform for students to manage their study activities in one place. It helps students manage subjects, create study tasks, track study time and monitor their overall progress.The Dashboard provides important information such as daily study goals, study time, subjects, day streak and progress. The Day Streak feature encourages students to study regularly by increasing the streak when they maintain their daily study activity.The Study Timer helps students conduct focused study sessions, while the My Progress section allows them to view their total study time, completed sessions and subject-wise progress. These features help students develop a consistent and organized study routine.The project also demonstrates the practical implementation of the MERN Stack, using React.js for the frontend, Node.js and Express.js for the backend, and MongoDB for storing application data.

Overall, StudySync provides a useful solution for students to plan, organize and monitor their studies effectively through a single web application

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